Rational Number Project

Initial Fraction Ideas Lesson 2.5: Overview	Materials
Students explore relationships among circle pieces, modeling and orally naming fraction amounts for: 1-half, 1-third, and 1-fourth.	• Fraction Circles for students and teacher • Student Page A, B

Teaching Actions

Warm Up

Using two different units, show the fraction 1-third. How are the two examples alike? How are the two examples different?

Large Group Introduction

1. Model fourths using 1 black, 1 yellow, and 1 brown as units.

2. Ask students to use the black circle as their unit and to find the color that partitions the black circle into equal parts. Ask: What fraction of the black circle is one blue piece?

3. Say: I want to show ¼ in another way. If one yellow piece is my unit or whole to be divided, what piece is ¼ of the yellow?

4. Say: Let's try to show ¼ in another way. If one brown is my unit, what piece is ¼ of the brown?

5. Ask: Can anyone explain why one blue, one gray, and one red can all be called 1-fourth?

6. End the development part of the lesson with a non-example. Show how 2 blues and 1 yellow cover the black circle. Pick up 1 blue and say that this piece is 1 of 3 parts of the circle so it is one-third of the circle. Ask: Is this true? If I wanted to know what part of the black circle 1 blue is, what must I do?

7. [Repeat showing 2 browns and 2 pinks covering

Comments

Students continue to explore unit fractions using the fraction circles extending to fourths.

While students are using the fraction circles, they are developing mental images for fractions that will later support their ability to order fractions.

Teaching Actions	Comments
the black circle. 1 pink does not equal 1-fourth]. **Small Group/Partner Work** 8. Student pages A & B present problems similar to ones presented in large group as well as problems within realistic contexts. Assign to students in pairs, as they are to answer the questions orally. **Wrap Up** 9. To assess the "big idea" in this lesson present the following scenario: Lianna said that 1 red piece is one-third; Rodrigo said 1 red is one-fourth. Who is correct? [Note that 1 red is one-third of the blue; 1 red is also one-fourth of the brown. Both Lianna and Rodrigo are correct once you know what unit they are comparing the red to].	

Translations

- Manipulative to verbal
- Pictures to verbal

Using two different units, show the fraction 1-third.

How are the two examples alike?

How are the two examples different?

The class will work together in groups or in pairs on these problems. Answers are to be given orally or by drawing a picture. On some of the problems children may want to use the fraction circles to help solve the problem.

1. The yellow piece is the unit.

   How many grays cover the yellow piece? _____

   1 gray is _____ of the yellow.

   (Say the word)

2. The blue piece is the unit.

   How many reds cover the blue piece? _____

   1 red is _____ of the blue.

   (Say the word)

3. The brown piece is the unit.

   How many reds cover the brown piece? _____

   1 red is _____ of the brown.

   (Say the word)

4. What color is 1-fourth of one black circle? _____

5. What color is 1-third of one black circle? _____

6. Draw a picture of a round pizza. Show on your drawing the pizza cut into 2 fair shares.

   Each fair share is _____ of the whole pizza.

   (Say the fraction name)

7. Here is a picture of a candy bar that someone has started to cut into pieces. Draw lines in to finish cutting the candy bar into equal parts.



The small piece of candy is _____ of the whole candy bar.
(Say the fraction name)

8. Karla has a large chocolate cookie. Draw a picture of Karla's cookie. Show on your drawing how she could divide the cookie into 4 fair shares.

Each part is _____ of the whole cookie.
(Say the fraction name)

9. William has a square pan of brownies. Draw a picture of William's pan of brownies. Show on your drawing how William could divide the pan of brownies into 3 fair shares.

Each part is _____ of the whole pan of brownies.
(Say the fraction name)